We solve the equation

$$C\frac{\partial T}{\partial t} + A + BT = QSa \quad .$$

And discretize it with a 1st order FD:

(1) Forward Euler: $C\dfrac{T^{n+1} - T^n}{\Delta t} = (QSa)^n - A - BT^n$

(2) Backward Euler: $C\dfrac{T^{n+1} - T^n}{\Delta t} = (QSa)^{n+1} - A - BT^{n+1}$

(3) Crank-Nicolson: $C\dfrac{T^{n+1} - T^n}{\Delta t} = \dfrac{1}{2}\left[(QSa)^{n+1} - BT^{n+1} + (QSa)^n - BT^n\right] - A$

To solve (2) we have: $T^0 = 5$

$$\left(\frac{C}{\Delta t} + B\right)T^{n+1} = \frac{CT^n}{\Delta t} + (QSa)^{n+1} - A$$

$$T^{n+1} = \left(\frac{CT^n}{\Delta t} + (QSa)^{n+1} - A\right) \Big/ \left(\frac{C}{\Delta t} + B\right)$$

Crank-Nicolson

$$T^{n+1} = \left(\frac{CT^n}{\Delta t} + \frac{1}{2}(QSa)^{n+1} + \frac{1}{2}(QSa)^n - \frac{1}{2}BT^n - A\right) \Big/ \left(\frac{C}{\Delta t} + \frac{1}{2}B\right)$$

If we have a nonlinear equation, e.g.

$$(4) \qquad \underbrace{4\pi r^2 H \varrho C \dot{T}(t)}_{\substack{\text{zeitabhängige} \\ \text{Wärmeenergie}}} = \underbrace{\pi r^2 S(1-\alpha)}_{\substack{\text{von der Erde} \\ \text{aufgenommene Strahlungsenergie}}} - \underbrace{4\pi r^{\varepsilon} \sigma T(t)^4}_{\substack{\text{reflektierte} \\ \text{Strahlungsenergie}}}$$

Eq. (1), (2) and (3) are trivial to solve, as it is linear. However, eq. (4) is nonlinear of the form

$$f(T^{n+1}) = 0$$

To obtain T^{n+1}, we can use Newton's method:

$$(5) \qquad f(T^{n+1}) = f(\tilde{T}^{n+1}) + \frac{\partial f}{\partial T^{n+1}}(\tilde{T}^{n+1}) \underbrace{(T^{n+1} - \tilde{T}^{n+1})}_{:= \Delta T} + \mathcal{O}(\Delta T^2)$$

And solve (5) iteratively, stating with $\tilde{T}^{n+1} = T^n$. Iteration $i+1$ reads

$$T_{i+1}^{n+1} = T_i^{n+1} - \underbrace{\frac{f(T_i^{n+1})}{\frac{\partial f}{\partial T^{n+1}}(T_i^{n+1})}}_{\Delta T}$$